Odorant-binding protein 18 contributes to tolerance-related responses to flonicamid and its metabolite in *Apis mellifera ligustica*

Peitong Chen[1,+], Ru Yan[1,2,+], Liangdong Wang[2], Jiaqi Huang[3], Yao Liu[4], Zhanyi Xu[1], Yuchen Zhang[1], Guoxing Chen[5], Jiali Qian[1], Keying Chen[5], Guonian Zhu[1], Huoqing Zheng[4], Mengli Chen[5,*], Yirong Guo[1,*]

[1] Institute of Pesticide and Environmental Toxicology, Zhejiang Key Laboratory of Biology and Ecological Regulation of Crop Pathogens and Insects, Ministry of Agriculture and Rural Affairs Key Laboratory of Molecular Biology of Crop Pathogens and Insect Pests, Zhejiang University, Hangzhou 310058, China;

[2] College of Life Sciences, Zhejiang University, Hangzhou 310058, China;

[3] College of Life Sciences, China Jiliang University, Hangzhou 310018, China;

[4] College of Animal Sciences, Zhejiang University, Hangzhou 310058, China;

[5] College of Advanced Agricultural Sciences, Zhejiang A&F University, Hangzhou, 311300, China.

[+] These authors contributed equally.

[*] Corresponding author *E-mail address*: yirongguo@zju.edu.cn (Yirong Guo).

cmlmeng@zafu.edu.cn (Mengli Chen)

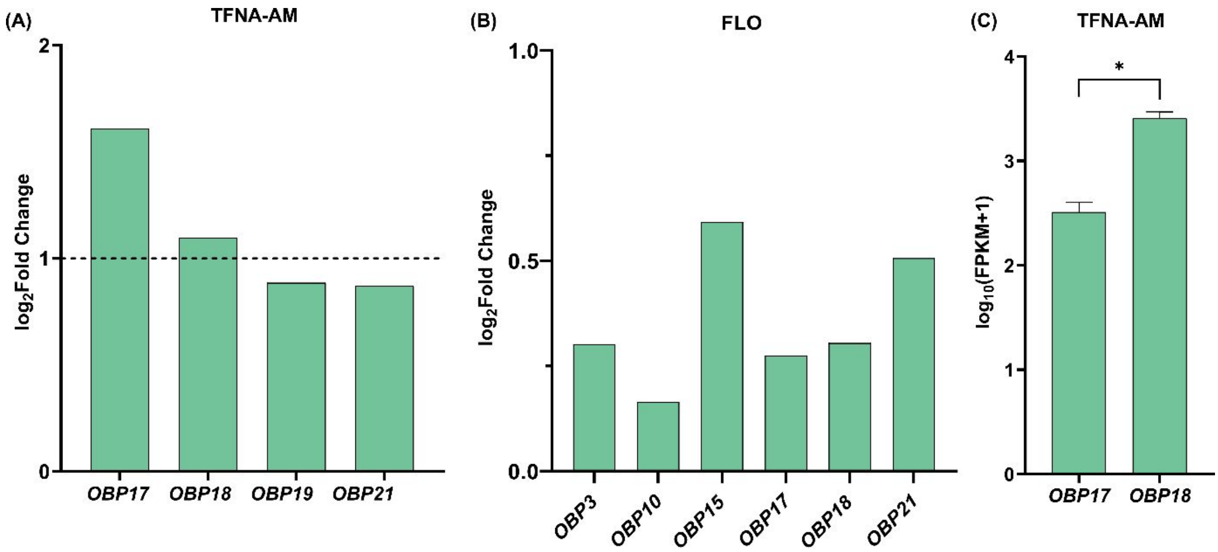

Figure S1. Differential expression of *AmOBPs* in honey bee antennae following TFNA-AM and FLO exposure. (A) Log_2(Fold Change) of four *OBPs* under TFNA-AM treatment. Genes with $|log_2$ (Fold Change) $| > 1$ were classified as differentially expressed genes. (B) Log_2(Fold Change) of four *OBPs* under FLO treatment. (C) Expression levels of *OBP17* and *OBP18* under TFNA-AM exposure, shown as log_{10}(FPKM + 1). Error bars represent the mean ± SE (Student's *t*-test, $*p < 0.05$).

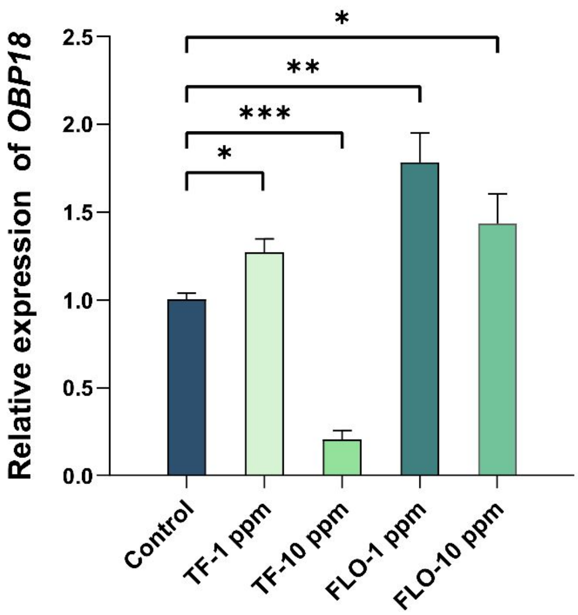

Figure S2. Expression of *AmOBP18* in legs after sublethal exposure to 1 ppm and 10 ppm of FLO and TF (TFNA-AM) exposure. (Student's *t*-test, $*p < 0.05$; $**p < 0.01$; $***p < 0.001$).

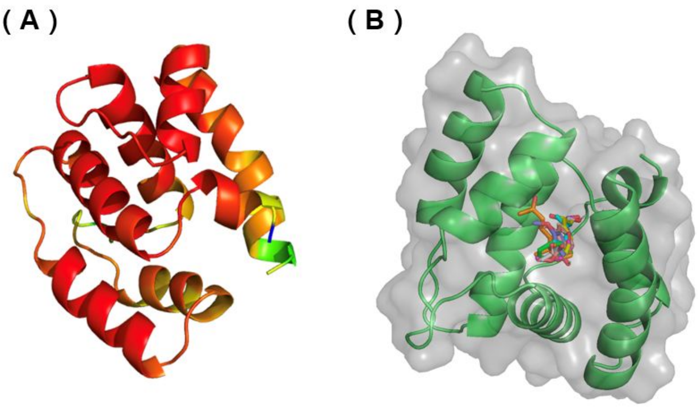

Figure S3. AlphaFold3-predicted structure of AmOBP18 and schematic illustration of molecular docking results. (A) The AF3-predicted structure of AmOBP18 is shown with confidence-based coloring, in which most residues are shaded in red or orange, indicating high prediction reliability. (B) Molecular docking of AmOBP18 with representative ligands. The protein is displayed as a green ribbon with a semi-transparent surface, and docked ligands are shown in stick representation within the binding cavity, illustrating the potential ligand-binding pocket of AmOBP18.

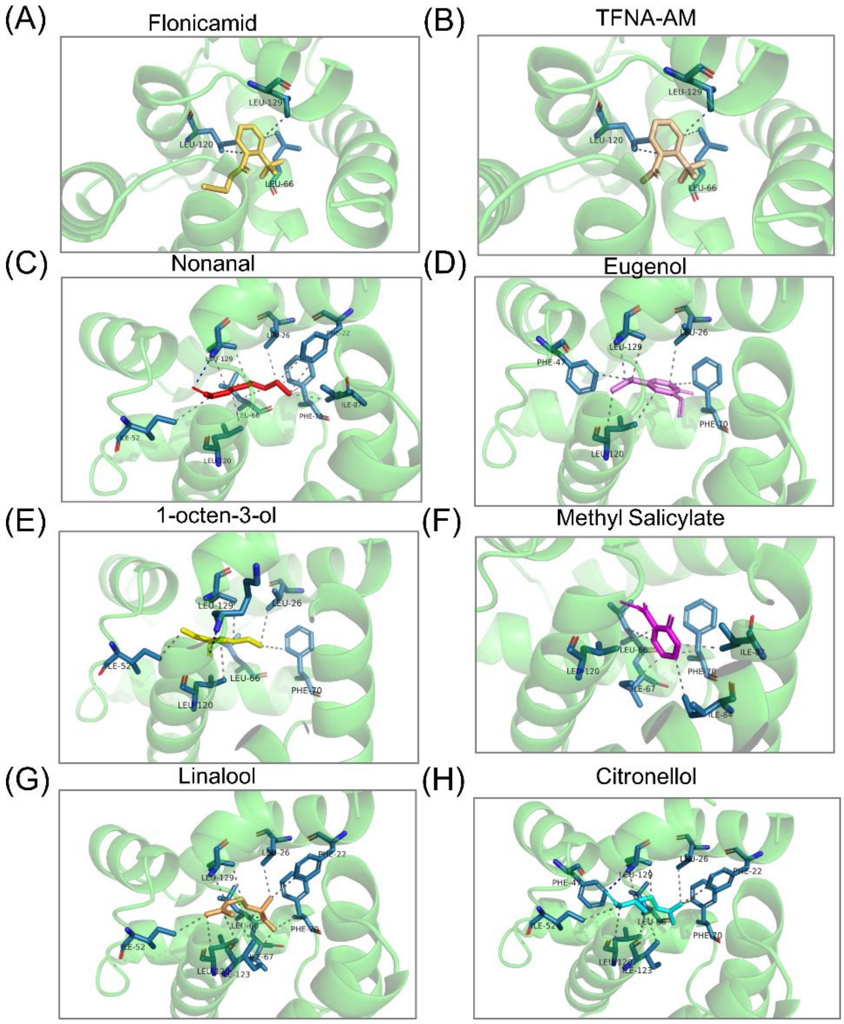

Figure S4. Enlarged views of the docking conformations of different ligands in the binding pocket of AmOBP18.

Table S1 Primers used in this study

Primer name	Sequence (5'-3')
For RT-qPCR analysis	
Actin-F	TTGTATGCCAACACTGTCCTTT
Actin-R	TGGCGCGATGATCTTAATTT
Am*OBP18*-qF	GCTATTTGCGTTTGCGTTG
Am*OBP18*-qR	TGATCAATGCTAGTTTCTA
For amplification	
AmOBP18-F	ATGCTTGAAGAATTTCAAATCGG
AmOBP18-R	TTAGCCACTTAACATTTCTTTCA
For dsRNA synthesis	
ds*EGFP*-F	TAATACGACTCACTATAGGGAAGTTCAGCGTGTCCGGC
ds*EGFP*-R	TAATACGACTCACTATAGGGCACCTTGATGCCGTTCTTC
ds*OBP18*-F	TAATACGACTCACTATAGGGGGAAATTTTAACGAAGTTGTG
ds*OBP18*-R	TAATACGACTCACTATAGGGTTTCAATGTTTTATATTTTCC

Table S2. Kinetic parameters, including association rate (*k*a), dissociation rate (*k*d), and binding affinity (K_D) of AmOBP18 for two insecticides and six floral ligands in a SPR analysis.

Ligands	*k*a(M^{-1} s^{-1})	*k*d (s^{-1})	K_D (μM)
TFNA-AM	5856	0.0009523	0.163
Methyl salicylate	3178	0.001835	0.577
Nonanal	2573	0.00303	1.18
Citronellol	2292	0.002799	1.22
1-octen-3-ol	2250	0.002765	1.23
Linalool	903.8	0.001511	1.67
FLO	2602	0.004789	1.84
Eugenol	655.3	0.01074	16.4

Table S3. Hydrophobic interactions of AmOBP18 with different ligands.

Ligand	Protein Residue Index	Distance/Å	Ligand Atom	Protein Atom
Flonicamid	LEU66	3.42	10	407
	LEU120	3.57	8	822
	LEU129	3.62	10	893
TFNA-AM	LEU66	3.49	5	405
	LEU120	3.47	4	820
	LEU129	3.63	5	891
Nonanal	PHE22	3.72	4	47
	PHE22	3.79	5	45
	LEU26	3.65	4	76
	ILE52	3.94	8	286
	LEU66	3.55	2	399
	LEU66	3.89	3	397
	PHE70	3.56	3	434
	PHE70	3.82	5	432
	ILE87	3.62	5	571
	LEU120	3.6	6	814
	LEU129	3.86	7	883
	LEU129	3.69	1	885
Eugenol	LEU26	3.72	6	79
	PHE47	3.51	7	249
	PHE70	3.53	6	437
	LEU120	3.69	9	814
	LEU120	3.6	2	817
	LEU129	3.67	8	886
	LEU129	3.75	7	888
1-octen-3-ol	LEU26	3.72	6	79
	PHE47	3.51	7	249
	PHE70	3.53	6	437
	LEU120	3.69	9	814
	LEU120	3.6	2	817
	LEU129	3.67	8	886
Methyl Salicylate	PHE22	3.77	12	49
	LEU26	3.85	12	78
	PHE47	3.38	3	248
	ILE52	3.92	3	288
	LEU66	3.73	7	401
	LEU66	3.71	9	399
Linalool	PHE22	3.85	2	49
	LEU26	3.62	2	78
	ILE52	3.6	9	288
	LEU66	3.88	5	401

	LEU66	3.97	11	399
	ILE67	3.61	12	410
	PHE70	3.66	12	433
	LEU120	3.77	9	813
	LEU120	3.66	8	816
	ILE123	3.64	10	838
	LEU129	3.73	7	885
	LEU129	3.44	6	887
Citronellol	PHE22	3.77	12	49
	LEU26	3.85	12	78
	PHE47	3.38	3	248
	ILE52	3.92	3	288
	LEU66	3.73	7	401
	LEU66	3.71	9	399
	PHE70	3.73	12	436
	LEU120	3.86	3	813
	LEU120	3.46	2	816
	ILE123	3.94	2	838
	LEU129	3.65	1	885
	LEU129	3.53	8	887

Table S4. Hydrogen-bond interactions of AmOBP18 with different ligands.

Ligand	Protein Residue Index	Distance/Å	Donor Angle	Ligand Atom
TFNA-AM	LEU129	3.20	149.62°	10[O_2] donor
1-octen-3-ol	LYS127	4.01	108.21°	5[O_3] acceptor
Citronellol	LEU129	3.03	154.34°	5[O_3] acceptor